DATASET COLLECTION REPORT

EmotionLens: Emotion Detector from Text

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Team: Team 13

Project Title: EmotionLens: Emotion Detector from Text

Dataset Source: Hugging Face

Dataset Name: DAIR-AI Emotion

Dataset Identifier: dair-ai/emotion

Configuration: split

Framework: Python and PyTorch

1. Introduction

Dataset collection is an important initial stage in developing an Artificial Intelligence model. The quality and organization of the dataset help the model learn patterns and make predictions.

For the EmotionLens project, the DAIR-AI Emotion dataset was obtained from Hugging Face. This dataset contains English text samples labelled with different emotions. It is used to develop a text-based emotion detection model that predicts the dominant emotion expressed in a message.

2. Objective of Dataset Collection

The main objectives of dataset collection are:

- To obtain a suitable dataset for text emotion classification.
- To collect labelled text samples for six emotion categories.
- To organize the dataset into training, validation, and testing sets.
- To examine the dataset structure and identify repeated texts and label inconsistencies.
- To prepare the dataset for training and evaluating a Transformer-based model.

3. Dataset Description

The project uses the DAIR-AI Emotion dataset available on Hugging Face.

Dataset source: https://huggingface.co/datasets/dair-ai/emotion

The dataset contains 20,000 text samples divided into three official splits.

Dataset split	Number of samples
Training set	16,000
Validation set	2,000
Testing set	2,000
Total	**20,000**

Each sample contains a text message and its corresponding emotion label.

The main dataset fields are:

- **Text:** The message or sentence used as model input.
- **Label:** The numerical category representing the emotion associated with the text.

4. Emotion Categories

The dataset supports six emotion categories:

1. Sadness
2. Joy
3. Love
4. Anger
5. Fear
6. Surprise

These categories are used as the target classes for the EmotionLens classification model.

5. Dataset Collection Procedure

The following procedure was followed to obtain and organize the dataset:

Step 1: Identify the dataset source

The DAIR-AI Emotion dataset was selected from Hugging Face because it provides text samples with emotion labels for text classification.

Step 2: Load the dataset

The dataset was loaded into the Google Colab environment using Python and the Hugging Face datasets library.

Step 3: Inspect the dataset

The available splits, row counts, column names, and data features were examined to confirm the expected dataset structure.

Step 4: Preserve the original dataset

The original training, validation, and testing splits were saved as separate Parquet files in the project's Google Drive folder.

Step 5: Audit the text samples

The dataset was examined for repeated texts, texts appearing across multiple splits, and cases where the same text had conflicting emotion labels.

Step 6: Save the dataset files

The source dataset files and audit results were stored in the permanent Team 13 project directory in Google Drive.

6. Dataset Quality Checks

During the dataset audit, the following results were recorded:

Audit item	Recorded result
Total dataset rows	20,000
Repeated-text rows	104
Unique repeated texts	52
Texts occurring across multiple splits	19
Texts with conflicting labels	51

These results indicate that some text samples are repeated or associated with inconsistent labels. Such cases should be documented and considered when interpreting model performance.

The official dataset splits were preserved, and no rows were deleted during the reported audit. Any further deduplication or conflict-handling policy should be documented before modifying the prepared dataset.

7. Dataset Storage Structure

The dataset and project files are organized in the following Google Drive directory:

My Drive/Applied AI/Team 13/

The relevant folder structure is:

- dataset/raw/ — Original source dataset files.
- dataset/processed/ — Prepared dataset files.
- artifacts/ — Saved model checkpoint, vocabulary, label names, and model configuration.
- results/ — Dataset audit reports, training history, test metrics, and evaluation results.

This structure helps maintain the dataset, model artifacts, and evaluation of evidence in a permanent project location.

8. Use of the Dataset

The collected dataset is used to train and evaluate the EmotionLens Transformer model.

The model learns from labelled text messages and predicts one of six emotion categories for a given input sentence. The training set is used to learn model parameters, the validation set is used for model selection, and the test set is reserved for final evaluation.

The model achieved the following results in the recorded test evaluation:

Evaluation metric	**Result**
Test accuracy	82.05%
Macro F1-score	0.7795
Weighted F1-score	0.8201

These results describe the model's performance on the project's test set. They do not imply that every individual emotion prediction will be correct.

9. Data Management and Limitations

The original source files should remain unchanged. Dataset preparation, preprocessing, and any future duplicate handling operations should be documented separately.

The project should preserve the official split assignments and ensure that vocabulary or other learned preprocessing components are fitted using training data only. Test labels and metrics should not be used to make model-selection decisions.

The dataset source should be credited in the project report. Its usage and redistribution terms should also be reviewed before distributing the dataset files outside the project team.

10. Conclusion

The dataset collection stage of the EmotionLens project obtained 20,000 labelled text samples from the DAIR-AI Emotion dataset on Hugging Face. The samples were organized into training, validation, and testing sets, and the original files and dataset audit results were saved in the Team 13 Google Drive project directory.

This dataset provides the foundation for developing a Transformer-based text emotion classifier. The next stages involve dataset preparation, model training, performance evaluation, and testing predictions on new text messages.